PARSHWANATH CHARITABLE TRUST'S
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Data Science

Campus Trace

Lost and Found Items within Campus

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Project Guide

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SDGs as per the UN framework:

Primary SDGs:

• **SDG 9: Industry, Innovation, and Infrastructure**
• **Why:** Your project builds a digital infrastructure for a traditional lost and found system within a campus. By integrating modern technologies such as React, Node.js, and MongoDB, it fosters innovation and creates a more efficient and scalable campus ecosystem.

• **SDG 11: Sustainable Cities and Communities**
• **Why:** The system improves organization and safety within the campus environment by enabling efficient tracking and recovery of lost items. It promotes a connected and cooperative community, contributing to a smarter and more sustainable campus.

SDGs as per the UN framework:

Secondary SDGs:

- **SDG 4: Quality Education**
- **Why:** The application supports students and staff by reducing the inconvenience caused by lost belongings, allowing them to focus better on academic activities and improving the overall educational experience.

- **SDG 12: Responsible Consumption and Production**
- **Why:** By increasing the chances of recovering lost items, the system reduces unnecessary replacement of belongings, thereby promoting reuse and minimizing waste.

1. Introduction

- Problem Identified :
    1. Students frequently lose items such as ID cards, wallets, keys, earphones, and notebooks within the campus.
    2. There is currently no **centralized digital platform** to report or track lost and found items.

- Solution Proposed :
    1. Develop a mobile/web application called **Campus Trace** where users can report lost or found items.
    2. The platform will help students **search, post and recover items** quickly within the campus community.

2. Objectives

1. **To design** and implement the user interface for a campus lost and found system.
2. **To create** a system that allows users to report lost and found items.
3. **To implement** a messaging system for communication between users.
4. **To implement** search and filter functionality for items.

3. Scope

1. **Can be applied in** college and university campuses to digitally manage and track lost and found items within the campus environment.
2. **Can be useful to** students, teachers, and campus staff to quickly report lost belongings and help others recover their items.
3. **Can be extended to** multiple campuses or institutions, allowing a wider network for reporting and searching lost items.
4. **Can help maintain** an organized and centralized database of lost and found items for easier tracking and management.

4. Feature /Functionality

Feature 1: User Registration and Login
Users can create an account and securely log in to access the system.

Feature 2: Report Lost Item
Users can post details of lost items including item name, description, location, and time.

Feature 3: Report Found Item
Users can upload information about items they have found on campus.

Feature 4: Search and Filter Items
Users can search items using keywords, category, or location.

Feature 5: Contact Owner/Finder
Users can communicate with the finder or owner to return the item.

5. Outcome of Project

1. **User can create an account and log in:**
   Students and staff can securely register and log into the application to access the lost and found services.
2. **User can report lost items:**
   Users can submit details of lost items including description, location, and time to help others identify them.
3. **User can report found items:**
   Anyone who finds an item on campus can post information about the item to help locate the rightful owner.
4. **User can search for lost or found items:**
   Users can browse or search the database to check if their lost item has been reported by someone.
5. **Users can contact the finder or owner:**
   The system enables communication between the person who found the item and the owner to arrange its return.

6. Technology Stack

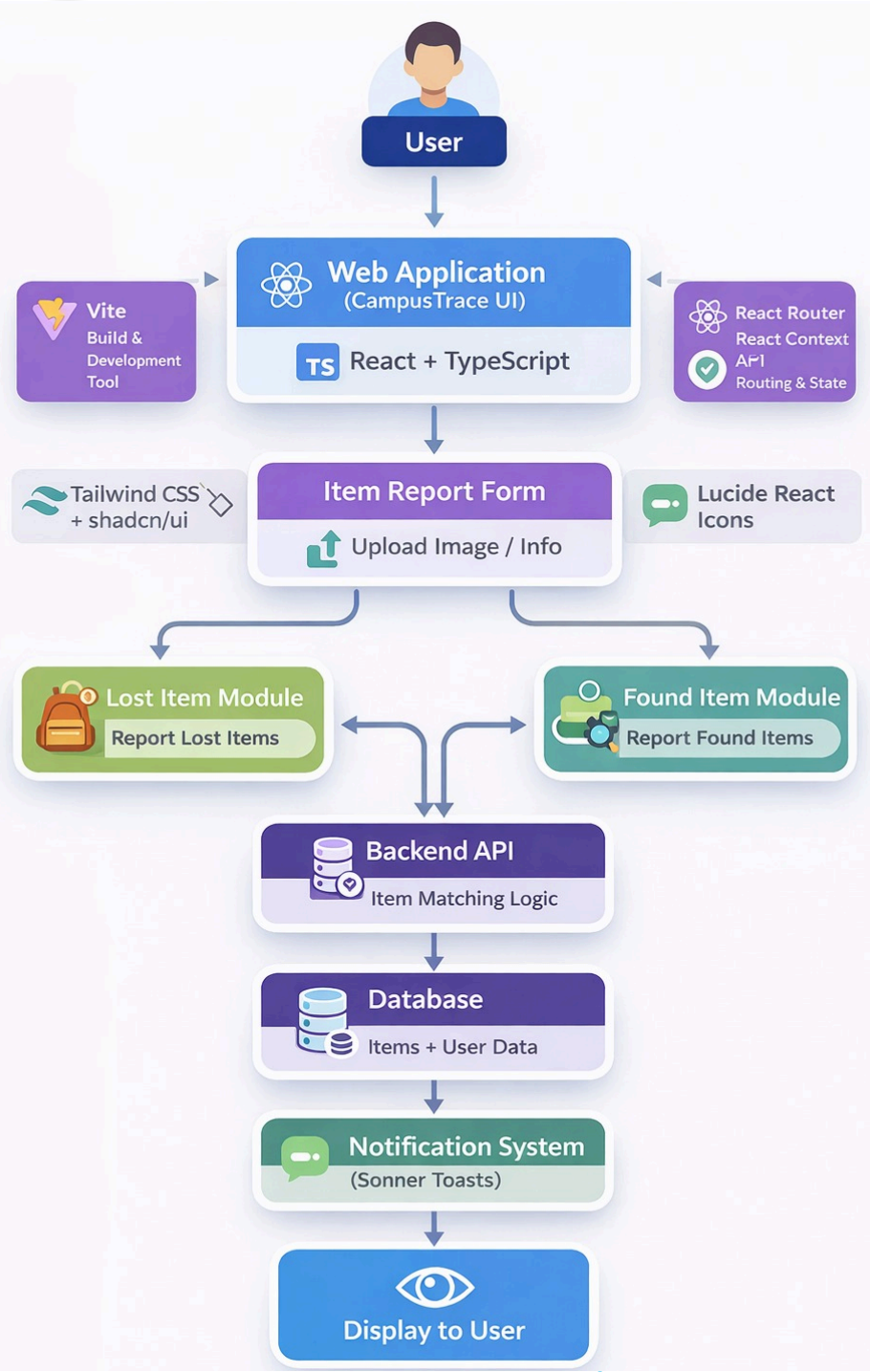
Frontend (GUI):

- **Language:** TypeScript
- **Framework:** React 18
- **Build Tool:** Vite
- **Styling:** Tailwind CSS with shadcn/ui components

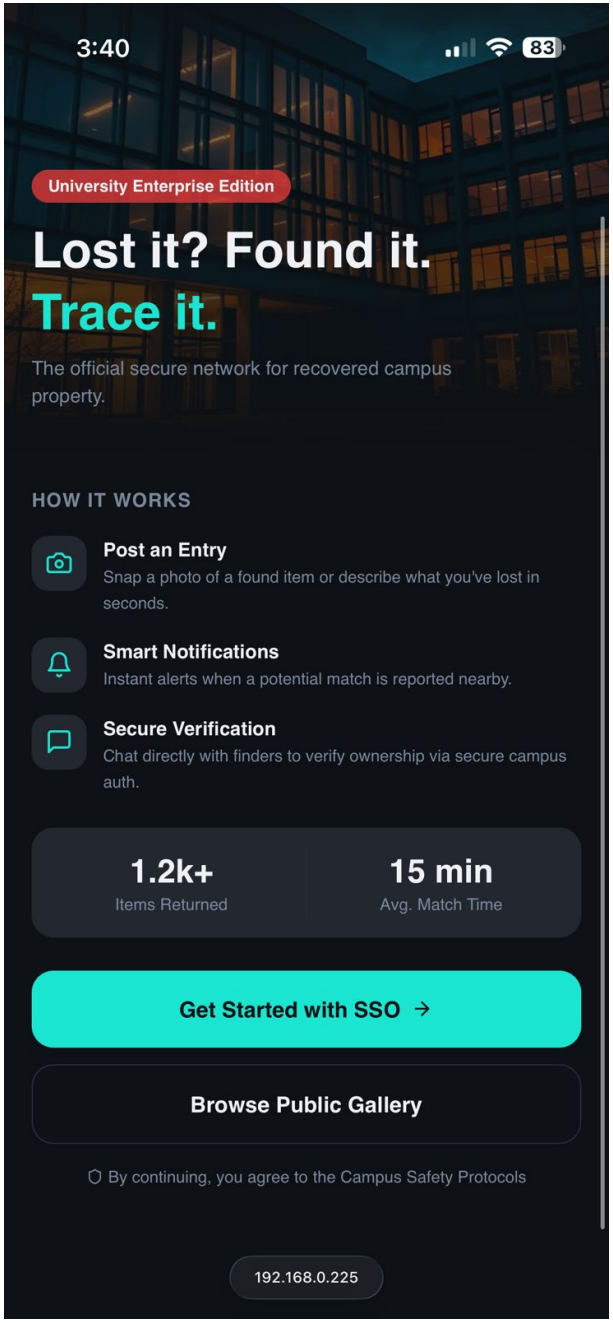
Application Features & Libraries:

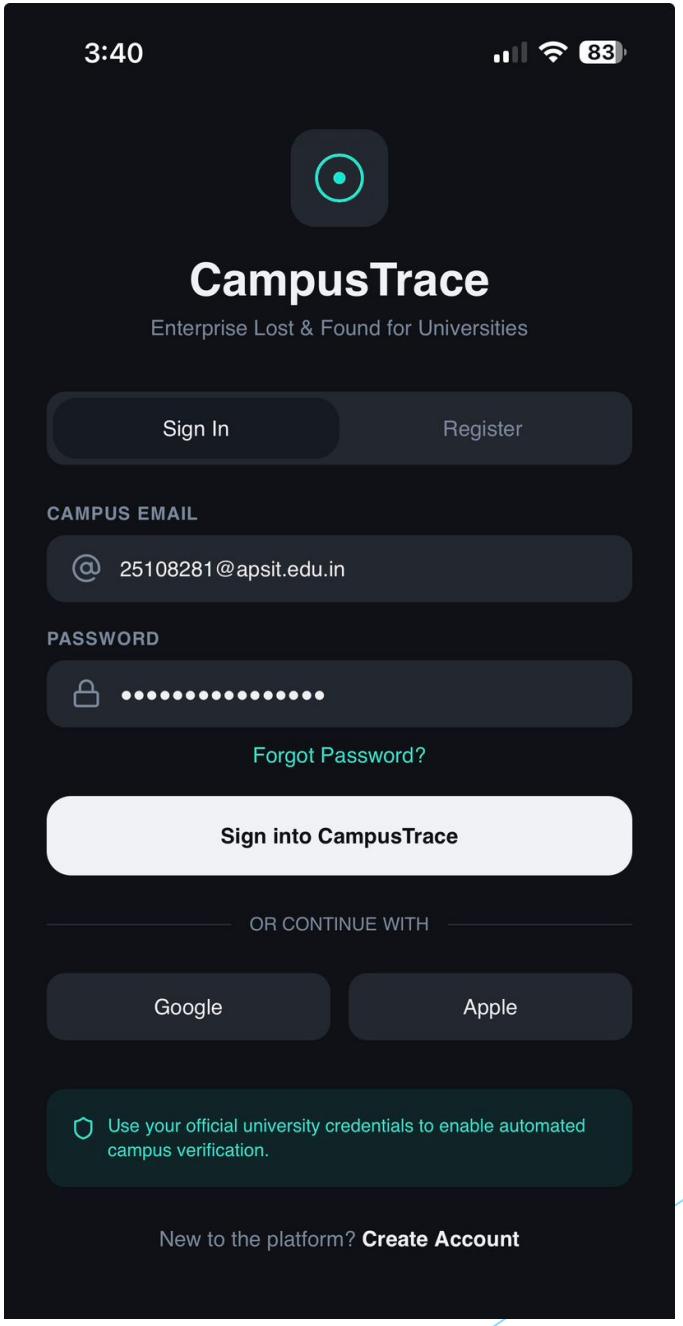
- **Routing:** React Router DOM (for navigation between pages)
- **State Management:** React Context API (for managing application data)
- **Icons:** Lucide React
- **Notifications:** Sonner (for toast notifications)

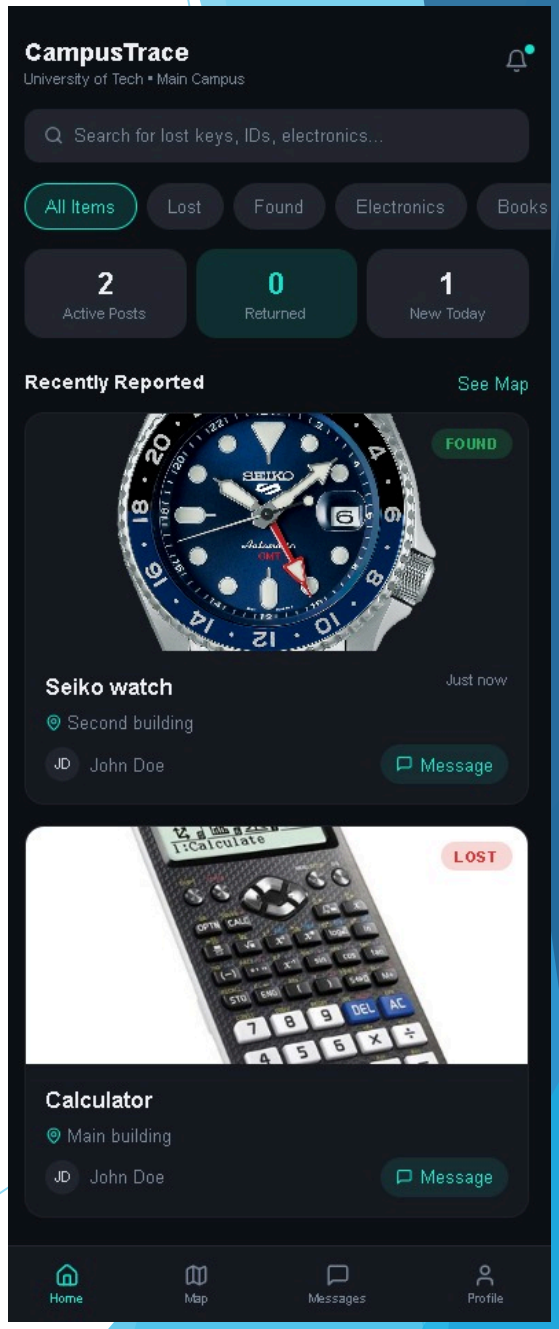
7. Block Diagram



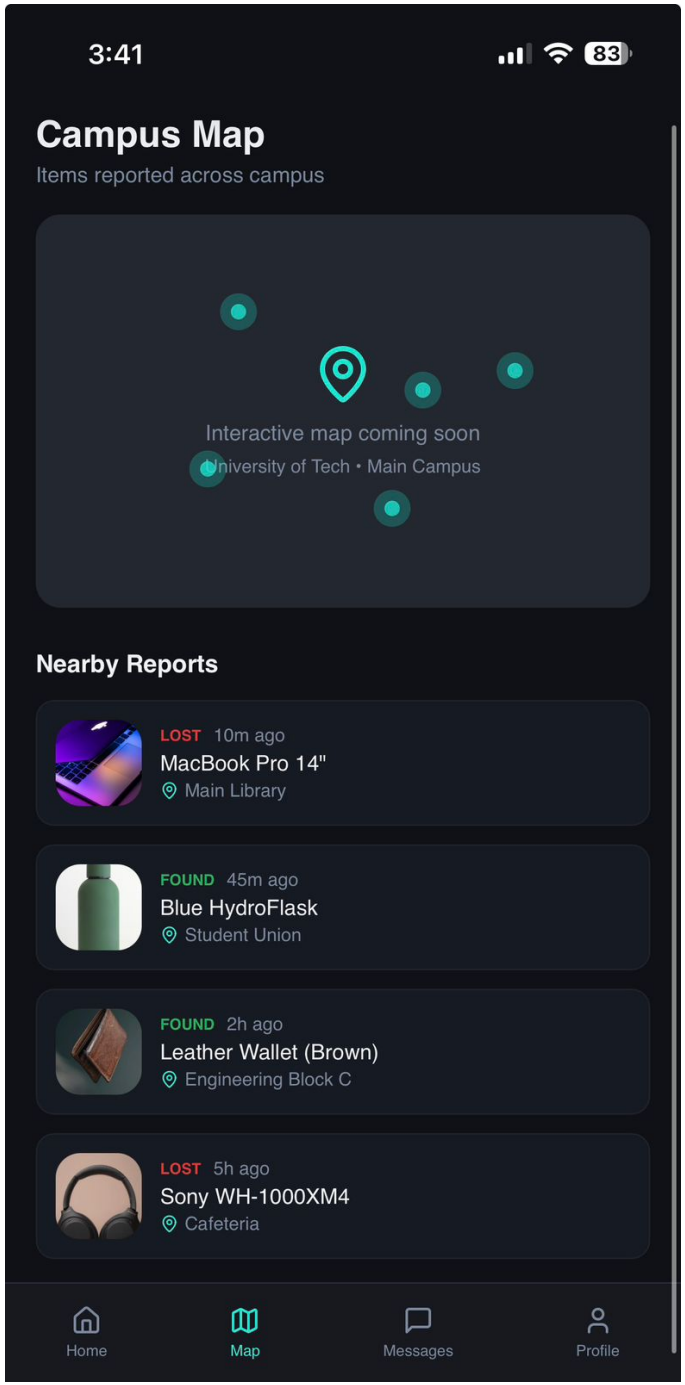
Application Output:

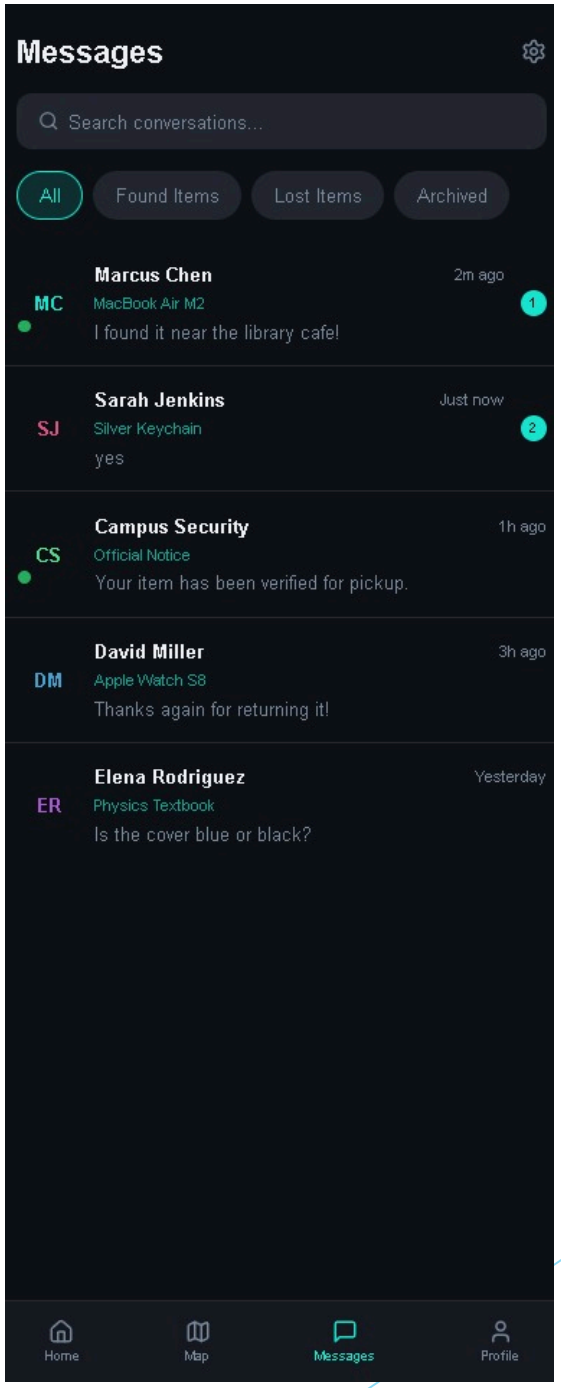


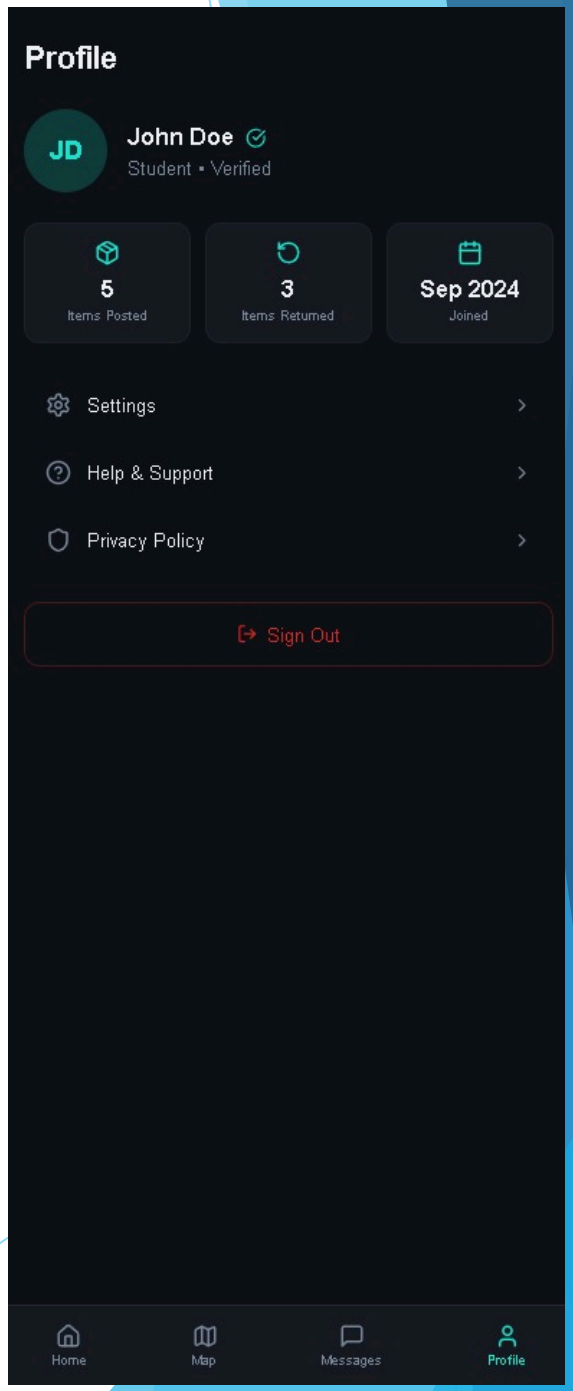




Application Output:







Thank You...!!